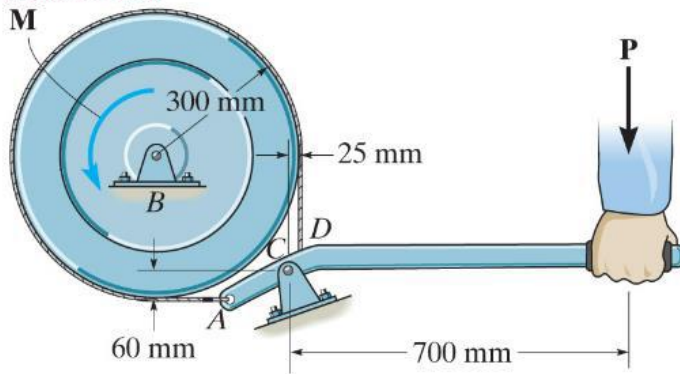
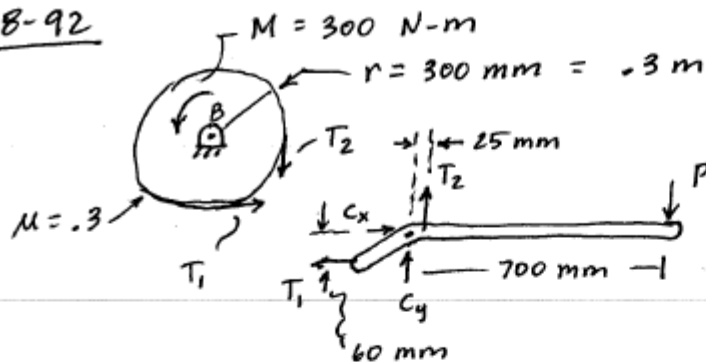


Oil Filter Wrench, Band Brake Example

Problems 8-92/93



8-92



From the drum:

$$+\circlearrowleft \sum M_B = 0; \quad 300 + T_1(0.3) - T_2(0.3) = 0$$

$$1000 = T_2 - T_1$$

Belt Friction:

$$T_2 = T_1 \cdot e^{\mu\beta}$$

$$= T_1 e^{(0.3)(1.5\pi)}$$

$$T_2 = 4.11 T_1$$

$$1000 = 3.11 T_1$$

$$T_1 = 321.4 \text{ N}$$

$$T_2 = 1321.4 \text{ N}$$

Handle:

$$+\circlearrowleft \sum M_C = 0; \quad 700 \cdot P + 60 T_1 - T_2(25) = 0$$

$$P = \frac{25 T_2 - 60 T_1}{700}$$

$$P = 19.64 \text{ N}$$